



A PROJECT REPORT

on

AI-Based Mental Health Stress Monitoring and Early Warning System Using NLP

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**BACHELOR'S DEGREE IN
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CERTIFICATE

This is to certify that the project entitled “**AI-Based Mental Health Stress Monitoring and Early Warning System Using NLP**” submitted by

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is a record of bona fide work carried out by her in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** at **KIIT Deemed to be University, Bhubaneswar**, during the academic year **2025–2026**, under my guidance.

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Place: Bhubaneswar

Project Guide

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ABSTRACT

Mental health issues such as stress, anxiety, and depression have become increasingly common among students and young adults due to academic pressure, lifestyle changes, and social factors. Early identification of such mental health concerns is challenging, as individuals often express their emotional state informally through text rather than seeking professional help at an early stage. This creates a need for intelligent systems that can assist in early stress monitoring and awareness.

This project presents an AI-based mental health stress monitoring and early warning system using Natural Language Processing (NLP) techniques. A pretrained transformer-based model, DistilBERT, is fine-tuned on a labeled mental health text dataset to classify user input into four categories: Normal, Anxiety, Depression, and Suicidal. The model achieves an accuracy of approximately 81% on the test dataset. To ensure ethical and responsible usage, the system incorporates a confidence-based filtering mechanism that withholds alerts when prediction confidence is low, thereby reducing false positives. Additionally, a lightweight explainability module highlights stress-related keywords influencing the prediction, improving transparency and user trust.

The trained model is deployed through a Streamlit-based web application that allows users to input text and receive real-time predictions along with confidence scores and explanations. The proposed system is intended as an early awareness and monitoring tool and not as a medical diagnostic solution. This project demonstrates the effectiveness of deep learning-based NLP models in supporting mental health awareness through responsible and interpretable AI systems.

Keywords: Natural Language Processing, Mental Health, Deep Learning, DistilBERT, Stress Detection

CHAPTER 1

Introduction

Mental health has emerged as a critical aspect of overall well-being, particularly among students and young adults. Academic pressure, competitive environments, social expectations, and lifestyle changes contribute significantly to increased levels of stress, anxiety, and depression. While severe mental health conditions often receive attention, early-stage stress and emotional imbalance frequently go unnoticed, which may later escalate into more serious concerns.

In recent years, individuals have increasingly expressed their thoughts and emotions through digital text platforms such as messages, online forums, and social media. These textual expressions often reflect an individual's mental state and can serve as valuable indicators for early stress detection. However, manually monitoring and interpreting such large volumes of unstructured text is impractical and subjective, highlighting the need for automated and intelligent solutions.

Advancements in Artificial Intelligence and Natural Language Processing (NLP) have enabled machines to understand and analyze human language with improved accuracy. Deep learning-based language models, particularly transformer architectures, have shown significant success in text classification tasks. These models are capable of capturing contextual and semantic relationships within text, making them suitable for identifying subtle emotional patterns associated with mental health conditions.

This project focuses on the development of an AI-based mental health stress monitoring and early warning system using NLP techniques. The proposed system utilizes a pretrained DistilBERT model, which is fine-tuned to classify user-provided text into four mental health categories: Normal, Anxiety, Depression, and Suicidal. The objective of the system is not to provide medical diagnosis, but to assist in early awareness and monitoring by analyzing textual inputs.

To ensure responsible usage, the system incorporates a confidence-based prediction mecha-

nism that avoids issuing alerts when the model's confidence is low, thereby reducing false positives and unnecessary concern. Additionally, an explainability component highlights stress-related keywords present in the input text, enabling users to better understand the basis of the prediction. The system is deployed as a web-based application to demonstrate real-time interaction and usability.

The remainder of this report is organized as follows: Chapter 2 discusses the fundamental concepts and related work relevant to the project. Chapter 3 outlines the problem statement, system requirements, and design methodology. Chapter 4 describes the implementation details and performance evaluation. Chapter 5 presents the standards adopted during development, and Chapter 6 concludes the report with future scope and enhancements.

CHAPTER 2

Basic Concepts and Literature Review

This chapter discusses the fundamental concepts and background required to understand the proposed mental health stress monitoring system. It introduces Natural Language Processing techniques, transformer-based deep learning models, and existing approaches to mental health text classification. These concepts form the theoretical foundation of the project.

2.1 Natural Language Processing

Natural Language Processing (NLP) is a branch of artificial intelligence that focuses on enabling machines to understand, interpret, and generate human language. NLP techniques are widely used to process unstructured textual data such as messages, reviews, and social media posts. Common NLP tasks include text preprocessing, tokenization, sentiment analysis, and text classification.

In mental health analysis, NLP plays a crucial role by extracting emotional and contextual information from textual expressions. Techniques such as lowercasing, removal of noise, and tokenization help transform raw text into a format suitable for machine learning models. NLP enables automated analysis of large volumes of text, which would otherwise be difficult to handle manually [1].

2.2 Deep Learning for Text Classification

Deep learning models have significantly improved the performance of text classification tasks compared to traditional machine learning approaches. Neural networks such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models were initially used to model sequential text data. However, these models often struggled with long-range dependen-

cies and computational efficiency.

Recent advancements have introduced transformer-based architectures that overcome these limitations by processing entire sequences in parallel. Deep learning models are capable of learning complex patterns and representations from text, making them suitable for identifying subtle emotional cues associated with mental health conditions.

2.3 Transformer Architecture

The transformer architecture is a deep learning model that relies on a self-attention mechanism to capture contextual relationships between words in a sentence. Unlike sequential models, transformers process all tokens simultaneously, enabling faster training and improved performance.

Self-attention allows the model to assign different importance weights to words based on their relevance within the context. This feature is particularly useful in mental health analysis, where the meaning of a word often depends on surrounding words. Transformer models have achieved state-of-the-art results in various NLP tasks, including sentiment analysis and text classification [2].

2.4 DistilBERT

DistilBERT is a lightweight version of the Bidirectional Encoder Representations from Transformers (BERT) model. It is designed to retain most of BERT's performance while reducing model size and computational complexity. DistilBERT achieves this through a knowledge distillation process, making it faster and more suitable for resource-constrained environments.

Due to its efficiency and strong contextual understanding, DistilBERT is widely used for text classification tasks. In this project, DistilBERT is fine-tuned on a labeled mental health dataset to classify textual inputs into predefined emotional categories. Its balance between performance and efficiency makes it appropriate for real-time applications [3].

2.5 Mental Health Text Classification

Mental health text classification involves categorizing textual data based on emotional or psychological states such as stress, anxiety, and depression. This task is challenging due to over-

lapping linguistic patterns between different mental health conditions. For example, anxiety and depression often share similar expressions, making precise classification difficult.

Several studies have explored the use of machine learning and deep learning models for mental health analysis using textual data. While these systems show promising results, ethical considerations such as false positives and lack of explainability remain concerns. The proposed system addresses these challenges by incorporating confidence-based filtering and explainability mechanisms to support responsible and transparent usage [4].

This chapter establishes the conceptual foundation required to understand the design and implementation of the proposed mental health stress monitoring system discussed in the subsequent chapters.

CHAPTER 3

Problem Statement and System Design

This chapter describes the problem addressed by the project, the objectives of the proposed system, functional and non-functional requirements, and the overall system design. It provides a structured overview of how the problem is analyzed and how the solution is planned before implementation.

3.1 Problem Statement

Mental health issues such as stress, anxiety, and depression often develop gradually and may go unnoticed during their early stages. Individuals frequently express their emotional states through informal textual communication rather than seeking professional help. Manual analysis of such unstructured textual data is subjective, time-consuming, and impractical at scale. There is a need for an automated system that can analyze textual input and provide early awareness regarding mental health stress levels in a responsible and ethical manner.

3.2 Objectives of the Project

The primary objectives of the proposed system are as follows:

- To analyze user-provided textual input using Natural Language Processing techniques.
- To classify text into predefined mental health categories such as Normal, Anxiety, Depression, and Suicidal.
- To develop an early stress monitoring system rather than a diagnostic tool.
- To incorporate confidence-based filtering to reduce false alerts.

- To provide explainability by highlighting stress-related keywords influencing the prediction.

3.3 Project Requirements

3.3.1 Functional Requirements

The functional requirements of the system include:

- Accept textual input from the user.
- Preprocess and tokenize the input text.
- Predict the mental health category using a trained deep learning model.
- Display prediction results along with confidence scores.
- Provide explanations when prediction confidence exceeds a predefined threshold.

3.3.2 Non-Functional Requirements

The non-functional requirements of the system include:

- Usability: The system should be simple and easy to use.
- Reliability: Predictions should be consistent for similar inputs.
- Ethical Consideration: The system should avoid false positives and clearly state that it is not a medical diagnostic tool.
- Performance: The system should provide real-time predictions with minimal delay.

3.4 System Design

The proposed system follows a modular design approach. The user provides textual input, which is processed using NLP techniques and passed to a fine-tuned DistilBERT model for classification. The model outputs prediction probabilities for each mental health category. A confidence-based filtering mechanism determines whether an alert should be displayed. An

explainability module highlights stress-related keywords influencing the prediction. Finally, the result is displayed to the user through a web-based interface.

3.4.1 System Architecture

The high-level architecture of the proposed system consists of the following components:

- User Interface for text input.
- Text Preprocessing and Tokenization Module.
- DistilBERT-Based Classification Model.
- Confidence-Based Decision Module.
- Explainability Module.
- Output Display Module.

The detailed implementation of these components is discussed in the next chapter.

SYSTEM ARCHITECTURE

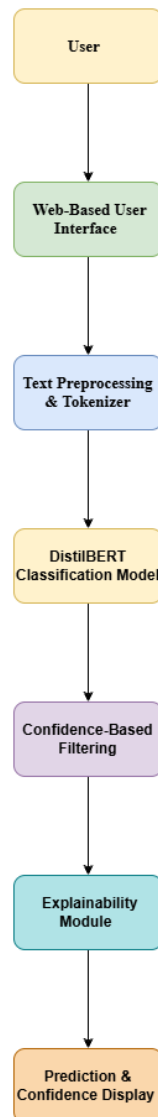


Figure 3.1: System Architecture of the Proposed Mental Health Stress Monitoring System

Figure 3.1 illustrates the high-level architecture of the proposed mental health stress monitoring system. The system accepts textual input from the user, processes it using NLP-based tokenization, and classifies the input using a fine-tuned DistilBERT model. A confidence-based filtering mechanism ensures responsible output, while an explainability module enhances transparency before displaying the final prediction to the user.

CHAPTER 4

Implementation and Results

This chapter describes the implementation details of the proposed mental health stress monitoring system. It explains the dataset used, preprocessing steps, model training methodology, and performance evaluation. The results obtained from the trained model are also discussed.

4.1 Dataset Description

The dataset used for this project consists of textual data related to mental health expressions. Each text sample is labeled into one of four categories: Normal, Anxiety, Depression, and Suicidal. The dataset is balanced, with an equal number of samples for each class, ensuring unbiased model training and evaluation.

The dataset was preprocessed to remove noise and inconsistencies, making it suitable for natural language processing tasks.

4.2 Data Preprocessing

Data preprocessing plays a crucial role in improving model performance. The following preprocessing steps were applied to the raw text data:

- Conversion of text to lowercase.
- Removal of URLs and special characters.
- Elimination of unnecessary whitespace.
- Removal of duplicate entries.
- Encoding of class labels into numerical format.

These steps ensure that the input text is clean and standardized before being fed into the deep learning model.

4.3 Model Implementation

The proposed system utilizes a pretrained DistilBERT model for text classification. DistilBERT is fine-tuned using the processed dataset to adapt its language representations for mental health classification. The dataset is split into training and testing sets in an 80:20 ratio.

The model is trained for two epochs using the AdamW optimizer. Training parameters such as batch size and sequence length are selected to balance computational efficiency and model performance, considering hardware constraints.

4.4 Confidence-Based Filtering and Explainability

To ensure ethical and responsible predictions, a confidence-based filtering mechanism is implemented. Predictions are displayed only when the model confidence exceeds a predefined threshold. If the confidence is low, the system marks the prediction as uncertain.

Additionally, an explainability module highlights stress-related keywords present in the input text. This improves transparency and helps users understand the basis of the model's predictions.

4.5 Performance Evaluation

The trained model is evaluated using standard classification metrics. The evaluation is performed on the test dataset, which constitutes 20% of the total data.

4.5.1 Accuracy

The proposed system achieves an overall classification accuracy of approximately 81.4% on the test dataset, demonstrating effective learning and generalization.

4.5.2 Precision, Recall and F1-Score

Accuracy alone may not fully represent the performance of a multi-class classification system. Therefore, additional evaluation metrics such as precision, recall, and F1-score were used to analyze the model's effectiveness for each mental health category.

Precision indicates the proportion of correct positive predictions, while recall measures the ability of the model to correctly identify all relevant instances. The F1-score represents the harmonic mean of precision and recall, providing a balanced measure of performance.

Table 4.1 presents the classification report obtained on the test dataset.

Table 4.1: Classification Report of the Proposed System

Class	Precision	Recall	F1-Score
Normal	0.95	0.84	0.89
Anxiety	0.70	0.92	0.80
Depression	0.77	0.60	0.67
Suicidal	0.87	0.90	0.88
Average	0.82	0.81	0.81

The results indicate strong performance across all classes, with particularly high recall for Anxiety and Suicidal categories, which is important for early stress detection. Lower recall for Depression reflects overlapping linguistic patterns with Anxiety, a known challenge in mental health text classification.

4.5.3 Confusion Matrix

A confusion matrix is used to analyze class-wise performance. The matrix shows strong classification performance for Normal, Anxiety, and Suicidal categories. Some overlap is observed between Anxiety and Depression classes due to similarity in linguistic expressions.

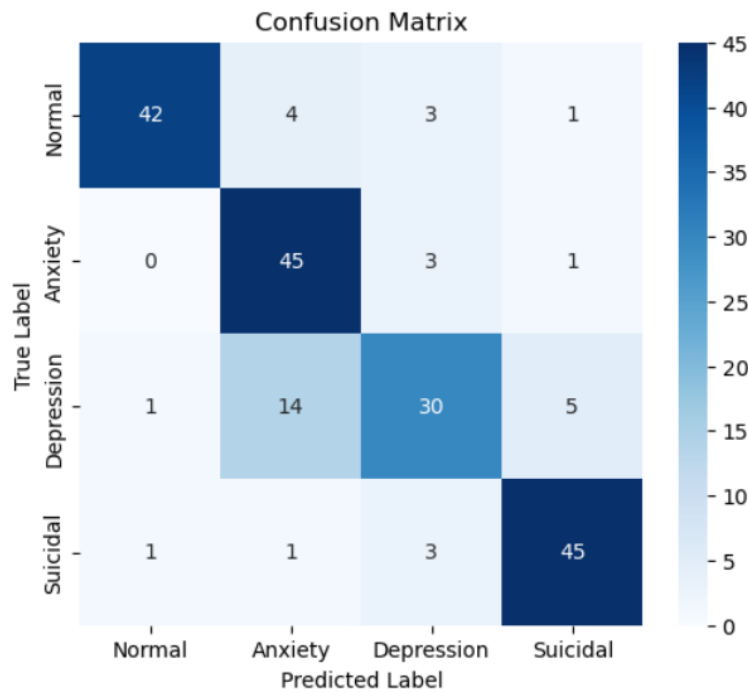


Figure 4.1: Confusion Matrix of the Proposed System

4.6 Web-Based Application

The trained model is deployed using a Streamlit-based web application. The application allows users to input text and receive real-time predictions along with confidence scores and explanations. The interface is designed to be simple and user-friendly.

Mental Health Stress Monitoring System

An AI-based early stress detection and monitoring tool

Enter your thoughts or feelings:

I feel like a burden to everyone around me. I don't like anything around me. I don't want to stay. I don't even know how I feel. I don't want to live.



Analyze



Result

Prediction: Depression

Confidence: 0.69

 Prediction confidence is low. Please provide more detailed input.

 This tool is for educational and early awareness purposes only. It is not a medical diagnostic system.

Figure 4.2: Web-Based Interface for Mental Health Stress Monitoring

The system includes a disclaimer stating that it is intended for early awareness and monitoring purposes only and should not be considered a medical diagnostic tool.

CHAPTER 5

Standards Adopted

This chapter describes the standards and best practices followed during the design, development, and testing of the proposed mental health stress monitoring system. Adhering to standard software engineering practices ensures reliability, maintainability, and ethical deployment of the system.

5.1 Design Standards

The system design follows a modular and layered architecture approach, where each component performs a specific function independently. This design methodology improves readability, scalability, and ease of maintenance. The architecture clearly separates user interaction, data preprocessing, model inference, decision-making, and output display.

Standard software design principles such as separation of concerns and modularity were adopted to ensure clarity and flexibility in system development.

5.2 Coding Standards

The implementation follows standard Python programming conventions. Meaningful variable names, modular functions, and proper indentation were used to improve code readability and maintainability. The code was structured in a logical manner to separate data preprocessing, model training, evaluation, and deployment components.

Well-documented libraries such as PyTorch, Hugging Face Transformers, and Streamlit were used to ensure robustness and compatibility with industry practices.

5.3 Testing Standards

The system was tested using functional testing techniques to verify that each module performs as expected. Black-box testing was applied by providing various textual inputs to the system and analyzing the corresponding outputs.

Performance testing was conducted using standard evaluation metrics such as accuracy and confusion matrix to assess the effectiveness of the classification model on unseen test data. Ethical testing considerations were also followed by ensuring that low-confidence predictions do not generate misleading alerts.

The adoption of these standards contributes to the overall quality, reliability, and responsible deployment of the proposed system.

CHAPTER 6

Conclusion and Future Scope

6.1 Conclusion

This project presented an AI-based mental health stress monitoring and early warning system using Natural Language Processing techniques. The system was designed to analyze user-provided textual input and classify it into mental health categories such as Normal, Anxiety, Depression, and Suicidal using a fine-tuned DistilBERT model. The primary objective of the project was to support early awareness and monitoring of mental health stress rather than provide medical diagnosis.

The implemented system demonstrated effective performance, achieving an overall classification accuracy of approximately 81.4% on the test dataset. The incorporation of a confidence-based filtering mechanism ensured responsible predictions by avoiding alerts for low-confidence cases. Additionally, the explainability module enhanced transparency by highlighting stress-related keywords influencing the model's decisions. The deployment of the system as a web-based application further demonstrated its practical usability and real-time interaction capability.

Overall, the project highlights the potential of deep learning-based NLP models in supporting mental health awareness through ethical, interpretable, and user-friendly AI systems.

6.2 Future Scope

Although the proposed system performs effectively, several enhancements can be considered for future development. The model can be extended to support multilingual text analysis, enabling broader accessibility across diverse user groups. Integration of voice-based input and speech-to-text conversion can further improve usability for users who prefer verbal expression.

Future work may also include training on larger and more diverse datasets to improve generalization and robustness. Advanced explainability techniques and continuous learning mechanisms can be explored to enhance interpretability and adaptability. Additionally, deploying the system on mobile platforms or integrating it with existing mental health support systems could increase its real-world impact.

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Individual Contribution

AI-BASED MENTAL HEALTH STRESS MONITORING AND EARLY WARNING SYSTEM USING NLP

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This project was carried out individually by **Aayushi** as part of the Bachelor of Technology program in Computer Science and Engineering.

The contributor was responsible for identifying the project problem, conducting background study, and selecting appropriate Natural Language Processing and deep learning techniques. The dataset used in the project was collected and preprocessed by the contributor, including text cleaning, labeling, and data preparation.

The contributor implemented the DistilBERT-based classification model, performed model training and evaluation, and analyzed the results using standard performance metrics. The confidence-based filtering mechanism and explainability module were also designed and implemented by the contributor to ensure ethical and transparent predictions.

Additionally, the contributor developed the web-based application for model deployment and prepared the complete project report, including documentation, diagrams, and result analysis. All stages of the project were completed independently under the guidance of the project supervisor.